

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 10%

Code of Conduct:

I Jagan R-192321102 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Jagan R

Rubrics for Evaluation

<i>Criteria</i>	<i>Weightage</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>	<i>Marks(100)</i>
<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>	
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>	
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>	
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>	
<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1. An image contains high-frequency noise. Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.

Answer :

When an image contains **high-frequency noise** (such as salt-and-pepper noise or random grainy noise), the most suitable frequency domain filter is the **Low Pass Filter (LPF)**.

Definition

A **Low Pass Filter (LPF)** allows **low-frequency components** (smooth regions of the image) to pass through while **blocking or reducing high-frequency components** (noise and sharp variations).

Working

1. Convert the image from the **spatial domain** to the **frequency domain** using the **Fourier Transform (FFT)**.
2. Apply a **Low Pass Filter** to remove or reduce the high-frequency components.
3. Keep the low-frequency components, which contain the main image information.
4. Convert the filtered image back to the spatial domain using the **Inverse Fourier Transform (IFFT)**.

Why Low Pass Filter is Used

- High-frequency noise appears as rapid intensity changes in the image.
- A Low Pass Filter suppresses these unwanted frequencies while preserving the important image content.
- Therefore, it is the most appropriate filter for reducing high-frequency noise.

Impact on Image Quality

Advantages:

- Reduces high-frequency noise effectively.
- Produces a smoother and cleaner image.
- Improves the visual appearance of noisy images.

Disadvantages:

- May blur image edges and fine details.
- Excessive filtering can reduce image sharpness.
- Some important high-frequency features may also be lost.

Examples of Low Pass Filters

- Ideal Low Pass Filter (ILPF)
- Butterworth Low Pass Filter (BLPF)
- Gaussian Low Pass Filter (GLPF)

Conclusion

A **Low Pass Filter** is the best choice for removing **high-frequency noise** in the frequency domain. It improves image quality by reducing noise, but if applied too strongly, it may also blur edges and reduce fine details. This makes it a trade-off between **noise removal** and **image sharpness**.

2. You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect. (6 Marks)

Answer:

To preserve low-frequency components while removing noise, the most suitable filtering technique is the Low Pass Filter (LPF).

Definition

A Low Pass Filter (LPF) is a frequency domain filter that allows low-frequency components to pass while blocking or reducing high-frequency components. Since most image noise is present in the high-frequency region, the filter removes noise while preserving the important image information.

Working

1. Convert the image from the spatial domain to the frequency domain using the Fourier Transform (FFT).
2. Apply a Low Pass Filter to suppress high-frequency noise.
3. Preserve the low-frequency components, which contain the overall brightness and smooth regions of the image.
4. Convert the filtered image back to the spatial domain using the Inverse Fourier Transform (IFFT).

How It Achieves the Desired Effect

- Low-frequency components represent the main structure and smooth areas of the image.
- High-frequency components mainly contain noise and sharp intensity changes.
- The Low Pass Filter removes the unwanted high-frequency noise while keeping the low-frequency information unchanged.
- As a result, the image becomes smoother with less noise. **Impact on Image Quality**

Advantages:

- Effectively removes high-frequency noise.
- Preserves the overall image structure.
- Produces a smoother and cleaner image.

Disadvantages:

- Can blur edges and fine details.
- Excessive filtering may reduce image sharpness.
- Some important high-frequency information may be lost.

Examples of Low Pass Filters

- Ideal Low Pass Filter (ILPF)
- Butterworth Low Pass Filter (BLPF)
- Gaussian Low Pass Filter (GLPF)

Conclusion

A Low Pass Filter (LPF) is the most suitable technique for preserving low-frequency components while removing high-frequency noise. It improves image quality by reducing noise and maintaining the important image content, although excessive filtering may cause slight blurring of edges.

3. An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components. (6 Marks)

Answer:

When an image requires edge enhancement in the frequency domain, the most suitable filter is the High Pass Filter (HPF).

Definition

A High Pass Filter (HPF) is a frequency domain filter that allows high-frequency components to pass while blocking or reducing low-frequency components. High-frequency components contain edges, fine details, and sharp intensity changes in an image.

Working

1. Convert the image from the spatial domain to the frequency domain using the Fourier Transform (FFT).
2. Apply a High Pass Filter to suppress low-frequency components.
3. Preserve the high-frequency components that contain edges and fine details.
4. Convert the filtered image back to the spatial domain using the Inverse Fourier Transform (IFFT).

Why High Pass Filter is Used

- Edges and fine details are represented by high-frequency components.
- The High Pass Filter removes smooth background information (low frequencies) and highlights sharp transitions.
- Therefore, it is the best filter for edge enhancement and image sharpening.

Impact on Frequency Components

- Low-frequency components: Reduced or removed.
- High-frequency components: Preserved and enhanced.
- As a result, edges become more visible and the image appears sharper.

Impact on Image Quality

Advantages:

- Enhances edges and fine details.

- Improves image sharpness.
- Makes object boundaries more visible.

Disadvantages:

- May amplify high-frequency noise.
- Excessive filtering can produce unnatural or noisy images.
- Smooth regions may lose their original appearance.

Examples of High Pass Filters

- Ideal High Pass Filter (IHPF)
- Butterworth High Pass Filter (BHPF)
- Gaussian High Pass Filter (GHPF)

Conclusion

A High Pass Filter (HPF) is the most suitable frequency domain filter for edge enhancement. It removes low-frequency components and preserves high-frequency components, resulting in a sharper image with enhanced edges. However, excessive use may also increase image noise.

4. A smooth version of an image is required. Describe and justify the frequency domain technique used for smoothing, explaining its operation and results. (6 Marks)

Answer:

When a **smooth version of an image** is required, the most suitable frequency domain technique is the **Low Pass Filter (LPF)**.

Definition

A **Low Pass Filter (LPF)** is a frequency domain filter that **allows low-frequency components** to pass while **blocking or reducing high-frequency components**. Since high-frequency components contain noise and sharp details, removing them produces a smoother image.

Working

1. Convert the image from the **spatial domain** to the **frequency domain** using the **Fourier Transform (FFT)**.
2. Apply a **Low Pass Filter** to suppress high-frequency components.

3. Preserve the low-frequency components, which contain the overall structure and smooth regions of the image.
4. Convert the filtered image back to the spatial domain using the **Inverse Fourier Transform (IFFT)**.

Why Low Pass Filter is Used for Smoothing

- High-frequency components represent edges, fine details, and noise.
- Low-frequency components represent the overall brightness and smooth regions.
- The Low Pass Filter removes high-frequency components, making the image appear smoother and reducing unwanted noise. **Results on Image Quality**

Advantages:

- Produces a smooth and visually pleasing image.
- Reduces high-frequency noise effectively.
- Preserves the overall structure and brightness of the image.

Disadvantages:

- Blurs edges and fine details.
- Excessive smoothing reduces image sharpness.
- Some important high-frequency information may be lost.

Examples of Low Pass Filters

- Ideal Low Pass Filter (ILPF)
- Butterworth Low Pass Filter (BLPF)
- Gaussian Low Pass Filter (GLPF)

Conclusion

A **Low Pass Filter (LPF)** is the most suitable frequency domain technique for **image smoothing**. It removes high-frequency components while preserving low-frequency components, resulting in a smoother image with reduced noise. However, excessive smoothing may blur edges and reduce fine details.

5. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.

Answer :

When an object is **clearly brighter than the background**, the most suitable segmentation technique is **Thresholding (Global Thresholding)**.

Definition

Thresholding is an image segmentation technique that separates the object from the background by comparing each pixel intensity with a predefined **threshold value (T)**.

- If the pixel intensity is **greater than T**, it is classified as the **object**.
- If the pixel intensity is **less than or equal to T**, it is classified as the **background**.

Working

1. Read the input image.
2. Choose a suitable threshold value (**T**). 3. Compare every pixel intensity with **T**.
4. Pixels with intensity $> T$ become **white (object)**.
5. Pixels with intensity $\leq T$ become **black (background)**.
6. The output is a **binary image** with the object clearly separated from the background.

Why Thresholding is Used

- The object is much brighter than the background.
- The difference in intensity makes it easy to separate the object using a single threshold value.
- It is simple, fast, and computationally efficient.

Effectiveness

Advantages:

- Simple and easy to implement.
- Fast processing with low computational cost.

- Effectively separates bright objects from a dark background.
- Produces a clear binary segmented image.

Disadvantages:

- Performance depends on selecting the correct threshold value.
- Not suitable if the image has uneven lighting.
- May not work well when object and background have similar intensities.

Applications

- Medical image segmentation.
- Document image processing.
- Object detection.
- Industrial quality inspection.

Conclusion

Thresholding is the best segmentation technique when the **object is brighter than the background**. It separates the object using a threshold value, producing a clear binary image. It is fast, simple, and highly effective when there is a significant intensity difference between the object and the background.

6. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.

Answer :

To separate the **foreground and background based on intensity**, the most suitable segmentation method is **Thresholding**.

Definition

Thresholding is an image segmentation technique that divides an image into **foreground** and **background** by comparing each pixel intensity with a **threshold value (T)**.

- If the pixel intensity is **greater than T**, it is classified as the **foreground**.
- If the pixel intensity is **less than or equal to T**, it is classified as the **background**. **Working**
 1. Read the input image.
 2. Select a suitable threshold value (**T**).

3. Compare the intensity of each pixel with **T**.
4. Pixels with intensity $> T$ are assigned to the **foreground** (white).
5. Pixels with intensity $\leq T$ are assigned to the **background** (black).
6. The output is a **binary image** that clearly separates the foreground from the background.

Why Thresholding is Used

- It separates objects based on differences in pixel intensity.
- It is simple, fast, and easy to implement.
- It works well when there is a clear contrast between the foreground and background.

Advantages

- Easy to implement.
- Fast and computationally efficient.
- Produces a clear binary image.
- Effective for images with good contrast.

Disadvantages

- Choosing the correct threshold value is important.
- Not suitable for images with uneven illumination.
- Performs poorly when foreground and background have similar intensity values.

Applications

- Object detection.
- Medical image analysis.
- Character recognition (OCR).
- Industrial inspection.

Conclusion

Thresholding is the most appropriate segmentation method for separating **foreground and background based on intensity**. It compares each pixel with a threshold value and creates a binary image, making it a simple, fast, and effective technique when the intensity difference is clear.

7. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.

Answer :

When an image requires **identification of object edges**, the most suitable edge detection method is the **Canny Edge Detection** algorithm.

Definition

Canny Edge Detection is a multi-stage edge detection technique that detects strong and continuous edges while reducing the effect of noise. It is one of the most accurate and widely used edge detection methods.

Working

1. **Noise Reduction:** Apply a **Gaussian filter** to remove noise from the image.
2. **Gradient Calculation:** Compute the intensity gradient using the **Sobel operator** to identify edge strength and direction.
3. **Non-Maximum Suppression:** Remove unnecessary pixels and keep only thin edges.
4. **Double Thresholding:** Classify edges as **strong, weak, or non-edges** using two threshold values.
5. **Edge Tracking by Hysteresis:** Connect weak edges to strong edges and remove false edges.

Why Canny Edge Detection is Used

- Detects edges accurately with less noise.
- Produces thin and continuous edges.
- Reduces false edge detection.
- Suitable for images requiring precise object boundary detection.

Effectiveness

Advantages:

- High edge detection accuracy.
- Good noise suppression.
- Produces thin and well-defined edges.
- Reduces false edge detection.

Disadvantages:

- More computationally expensive than simple edge detectors.
- Requires proper selection of threshold values.
- Slightly slower due to multiple processing stages.

Applications

- Object detection.
- Medical image analysis.
- Face recognition.
- Computer vision and robotics.
- Image segmentation.

Conclusion

Canny Edge Detection is the most suitable method for identifying **object edges**. It uses noise reduction, gradient calculation, non-maximum suppression, double thresholding, and edge tracking to produce accurate and continuous edges. It provides high-quality edge detection with minimal noise, making it one of the most effective edge detection techniques.

8. Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.

Answer :

When **adjacent pixels have similar intensity values**, the most suitable segmentation technique is **Region Growing**.

Definition

Region Growing is a region-based image segmentation technique that groups **neighboring pixels with similar intensity values** into a single region. It starts from a selected **seed point** and expands the region by adding similar adjacent pixels.

Working

1. Select one or more **seed points** in the image.
2. Compare the intensity of neighboring pixels with the seed pixel.
3. If the neighboring pixel has a similar intensity, add it to the region.

4. Repeat the process until no more similar neighboring pixels remain.
5. The final output is a segmented region containing pixels with similar intensity.

Why Region Growing is Used

- It groups connected pixels with similar intensity.
- It produces accurate object boundaries.
- It is suitable for images where neighboring pixels belong to the same object.

Effectiveness

Advantages:

- Produces accurate segmentation.
- Preserves connected regions.
- Simple and easy to implement.
- Works well for images with uniform intensity.

Disadvantages:

- Depends on the selection of seed points.
- Sensitive to noise.
- Can be slow for large images.
- May produce incorrect regions if intensity varies greatly.

Applications

- Medical image segmentation.
- Satellite image analysis.
- Object detection.
- Industrial image inspection.

Conclusion

Region Growing is the most suitable segmentation technique for grouping **adjacent pixels with similar intensity values**. It starts from a seed point and expands by adding neighboring pixels with similar intensity,

resulting in accurate and connected image regions. It is effective for images with uniform intensity but depends on proper seed selection.

9. Detected edges in an image appear broken. Explain and justify a method to improve edge continuity, including its role in post-processing.

Answer :

When the **detected edges in an image appear broken**, the most suitable method is **Morphological Closing** (Dilation followed by Erosion).

Definition

Morphological Closing is a post-processing technique used to **connect broken edges, fill small gaps, and produce continuous object boundaries**. It consists of **Dilation** followed by **Erosion**. **Working**

1. Start with the edge-detected image.
2. Apply **Dilation** to expand the edges and connect nearby broken edge segments.
3. Apply **Erosion** to remove extra pixels added during dilation while keeping the connected edges intact.
4. The output is an image with **continuous and smoother edges**.

Why Morphological Closing is Used

- Connects broken or disconnected edges.
- Fills small gaps and holes in object boundaries.
- Improves the quality of edge detection.
- Acts as an important **post-processing** step after edge detection.

Role in Post-Processing

- Refines the output of edge detection algorithms such as **Canny**, **Sobel**, or **Prewitt**.
- Removes small breaks and irregularities.
- Produces complete and connected object boundaries for further image analysis.

Effectiveness

Advantages:

- Connects broken edges.
- Fills small gaps and holes.
- Produces smooth and continuous boundaries.
- Improves object detection and segmentation.

Disadvantages:

- Excessive closing may merge nearby objects.
- Choice of structuring element affects the result.
- May slightly alter object shape.

Applications

- Object recognition.
- Medical image processing.
- Character recognition (OCR).
- Industrial image inspection.

Conclusion :

Morphological Closing is the most suitable method for improving **edge continuity**. By applying **dilation followed by erosion**, it connects broken edges, fills small gaps, and produces continuous object boundaries. It is an effective **post-processing** technique that improves the quality of edge detection results.

10. You need to extract object outlines from an image. Describe and apply an appropriate technique, explaining how it identifies boundaries effectively.

Answer :

To extract object outlines from an image, the most suitable technique is **Edge Detection using the Canny Edge Detection algorithm**.

Definition

Canny Edge Detection is an edge detection technique that identifies the **boundaries (outlines)** of objects by detecting sudden changes in pixel intensity. It produces clear, thin, and continuous edges. **Working**

1. Apply a **Gaussian filter** to remove image noise.
2. Calculate the intensity gradient using the **Sobel operator**.
3. Apply **Non-Maximum Suppression** to keep only the strongest edge pixels.
4. Use **Double Thresholding** to classify pixels as strong edges, weak edges, or non-edges.
5. Perform **Edge Tracking by Hysteresis** to connect valid edges and remove false edges.
6. The output is an image showing only the **object outlines**.

Why Canny Edge Detection is Used •

- Detects object boundaries accurately.
- Produces thin and continuous outlines.
- Removes false edges caused by noise.
- Provides better results than simple edge detection methods.

Effectiveness

Advantages:

- High edge detection accuracy.
- Produces clear and continuous object outlines.
- Reduces noise effectively.
- Minimizes false edge detection.

Disadvantages:

- More computationally expensive than basic edge detectors.
- Requires proper threshold selection.

- Slightly slower due to multiple processing stages.

Applications

- Object recognition.
- Medical image analysis.
- Face detection.
- Computer vision.
- Image segmentation.

Conclusion

Canny Edge Detection is the most suitable technique for **extracting object outlines**. It detects boundaries by identifying sudden intensity changes while removing noise and false edges. The result is a clear, thin, and continuous outline of objects, making it one of the most effective edge detection techniques.